



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of CSE (Artificial Intelligence and Machine Learning)

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: I Semester – B.E. CSE (AIML)

Course Code & Title: CS25C01 Computer Programming: C

Name of the Faculty member (s): Dr. R. Venkatesh, ASP & Head /CSE (AIML)

Date of Implementation : 26.09.2025

Innovative Practice Description

- **Unit / Topic:** Module II / While, do-while loop
- **Course Outcome:** CO2
- **Activity Chosen:** Think-Pair-Share
- **Justification:**

The topic “While and Do-While Loops” is essential in programming, as these looping constructs form the foundation for iterative logic, control flow, and algorithmic thinking. Understanding the difference between entry-controlled (while) and exit-controlled (do-while) loops is crucial for writing efficient programs.

The Think–Pair–Share (TPS) strategy enables students to analyze logic, compare loop behaviour, correct errors, and apply loops in real-world problem-solving. This interactive activity enhances reasoning, peer learning, and communication skills.

Time Allotted for the Activity:

- **Procedure of the Implementation:**
 - Collaborative, active learning strategy, in which students work on a problem posed by instructor, first individually (Think), then in pairs (Pair) or groups, and finally together with the entire class (Share).
 - T (Think): Teacher asks a specific question about the topic. Students “think” about what they know or have learned, and come up with their own individual answer to the question. [Takes 1-5 Minutes].
 - P (Pair): Teacher asks another question, related to the previous one, that is suitable to deepen the students’ understanding of the topic. Each student is paired with another student. They share their thinking with each other and proceed with the task. [Takes 2-5 Minutes].
 - S (Share): Students share their thinking (or solution) with the entire class. Teacher moderates the discussion and highlights important points. [Takes 05-10 minutes].

Details of the Implementation

- The TPS activity was conducted to help students understand the behaviour, use cases, and differences between while and do-while loops in C programming. The session was conducted in three phases:

The instructor introduced the topic and explained why loops are required in programming.

Short examples of real-world repetitive tasks (ATM PIN attempts, menu-driven programs, sensor reading loops) were presented.

Code snippets highlighting the difference between:

- Entry-controlled loop: while
- Exit-controlled loop: do-while, were shown to provide context.
- **Think Phase (1 - 5 minutes):**

The instructor posed a question such as:

“What will be the output of a while loop if the condition is false initially?”

“Why does a do-while loop execute at least once?”

“Predict the output of the following code fragment:”

```
int i = 5;
do {
    printf("%d ", i);
    i++;
} while(i < 5);
```

Each student reflected on the logic individually and wrote down their answer.

Pair Phase (2 - 5 minutes):

- Students paired up to compare their logic and reason out differences in predicted outputs.
- They discussed the importance of:
 - loop initialization
 - condition checking
 - infinite loop possibilities
 - increment/decrement placement
- Slow learners clarified doubts and corrected misconceptions with peer help.

Share Phase (5 -10 minutes):

- Each pair shared their answers or corrected code with the class.
- Presentations included:
 - explanation of loop flow
 - reasoning for output differences
 - identifying errors such as missing semicolon, misplaced increments, or infinite loop traps
- The instructor moderated, validated the reasoning, and demonstrated the flow using a trace table.
- **Figure 1** illustrates students participating in the TPS activity for “While and Do-While Loops”.
- The session concluded with a whole-class discussion on the final answers and common mistakes in loop-based programs.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO2	3	3	3	2	3	2

(1 – Low

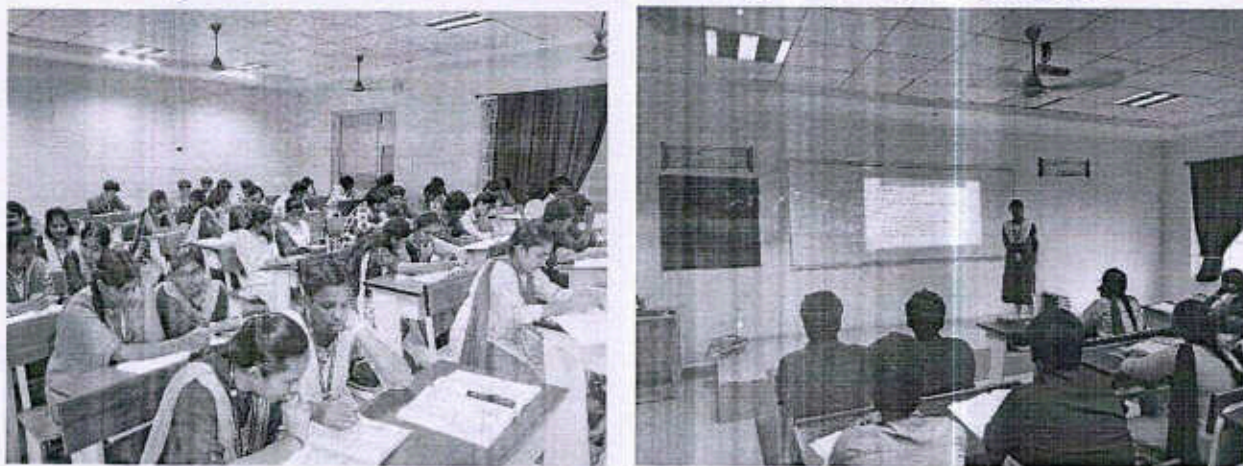
2 – Moderate

3 – High)

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
Justification for correlation	Students interpret loop behaviour and apply analytical skills to predict outputs.	Students select appropriate loop constructs to solve problems.	Students develop and modify programs using while/do-while loops.	Team collaboration improves during Pair and Share phases.	Presentation and explanation skills are strengthened.	Logical thinking and problem-solving using fundamental programming constructs are enhanced.

- **Images / Screenshot of the practice:**

Figure 1 – Students Think and Share their Thoughts and Topic Presentation



(Figure 1 – Students Think and Share their Answers on While/Do-While Loops - Sasika S Shared the reasoning)

- **Reflective Critique:**

- ❖ *Feedback of practice from students and other stakeholders:*

- Students actively participated and enjoyed the code-based reasoning tasks.
- They appreciated seeing different loop outputs and debugging small programs.
- Peer learning benefitted bright students as well as average learners.

- ❖ *Benefit of the practice:*

- Students were actively engaged in analyzing program flow.
- Peer explanation helped clarify the difference between while and do-while loops.
- Encouraged logical thinking and prediction of outputs.
- Made the class interactive and supportive.
- Ensured participation from most students in the learning process.

❖ **Challenges faced in implementation:**

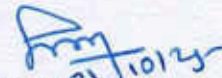
- Students unfamiliar with TPS needed initial guidance to follow the activity format.
- Some slow learners hesitated to participate actively without encouragement.
- Noise levels increased during the Pair phase and required moderation.
- Maintaining focus and keeping every student on-task required effort.
- The activity consumed more time than traditional lecturing.

References:

- <https://www.readingrockets.org/strategies/think-pair-share>
- <https://www.prodigygame.com/blog/active-learning-strategies-examples/>
- GeeksforGeeks – While and Do-While Loops in C
- Tutorialspoint – C Programming Loops
- adlit.org, “Adolescent Literacy”, 2019. [Online]. Available: <http://www.adlit.org/strategies/23277/> [Accessed: 23.01.2020]
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- prodigygame.com, “active-learning-strategies”, 2019. [Online].
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Signature of Faculty Member



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